

Automatic Recognition of Gesture Qualities in Speech Delivery in VR -Work-in-Progress

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ABSTRACT

Gesture use during public speaking can be analyzed as an aspect of speaker performance as well as an indicator of speaker emotions. Recognizing gesture quality rather than identifying specific gestures presents an underexplored challenge compared to traditional gesture recognition in virtual reality (VR). To that end, we use a VR headset and controllers to create a database of 162 five-second-long gestures. Next, we ask ten judges to evaluate the quality of gestures in three dimensions, i.e., dynamics, range, and between-hand distance, on a low-medium-high scale with average Fleiss Kappa 0.39 inter-rater agreement. A comparison of various classifiers revealed that the Support Vector Machine (SVM) yielded the best results, achieving classification accuracy of 70-85% for each quality dimension.

Index Terms: Gesture—Speech—Training—Machine learning; Automatic feedback—Public speaking

1 INTRODUCTION

The public speakers' gestures analysis is crucial in understanding and evaluating their overall performance [21]. Non-verbal communication, including gestures, is a powerful component of human interaction, conveying emotions and emphasis, helping to clarify verbal messages [18]. Studies have shown that effective use of gestures can enhance the speaker's persuasiveness, credibility, and audience engagement [17]. Well-timed and expressive gestures can create a sense of rapport between the speaker and the audience, fostering a more engaging atmosphere [14]. Furthermore, synchronising verbal and non-verbal elements, such as gestures, facial expressions, and body posture, contributes to a more coherent and impactful message, ultimately improving the overall communication experience [10].

Various gestural styles can be employed to enhance the message's effectiveness and clarity. The common styles include *descriptive gestures*, which involve using hand movements to illustrate a concept or idea; *emblematic gestures*, which are culturally specific signs or symbols; and *regulatory gestures*, which are employed to guide and control the conversation, such as raised hands to signal a pause or request for silence. Additionally, *affective gestures* express emotions and attitudes, while *interactive gestures* engage the audience. It is essential to balance these gestural styles and adapt them to the presentation's situational contexts, as the appropriate gesturing can increase the speaker's credibility and audience engagement [11, 17]. Therefore, analyzing, understanding and providing feedback on a speaker's gestures is vital to ensure effective communication.

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Moreover, when automated, such feedback can be, for instance, helpful in public speaking and voice emission training in front of a virtual audience during relaxation exercises [12]. The novelty of our approach lies in addressing this specific context of gesture analysis through a user study. Here, we limited the database to gestures from only one speaker and will scale to a database of multiple speakers in the near future as our approach showed significant promise for further expansion.

2 METHODS

2.1 Metrics

To measure the quality of gestures, we selected the following features and a 3-point ordinal scale to measure their intensity: *gesture dynamics*: low, medium, high; *distance between hands*: low, medium, high; *gesture range* (i.e., the distance of hands from the body): low, medium, high. The metrics were selected following consultation with public speaking trainers. They also appear in the literature considering human hand kinematics and dynamics in the communication and ergonomic aspects (i.a., [5]). Although the dimensions are not orthogonal, they can be universally measured using controller tracking with the most popular VR headsets available on the market, including those with external and inside-out tracking.



Figure 1: Example of a single gesture (a video frame).

2.2 Data Acquisition and Labeling

In our experiments, we used the Oculus Quest headset together with both controllers (see Fig 1). Each data record comprises 15 measurements, including the coordinates of the headset and controllers' positions (x, y, z) as well as acceleration along the X-, Y- and Z-axes for both controllers. Data was recorded with a sampling rate of 25

Hz, and each entry was accompanied by a video recording of the user for manual data labelling.

As the adopted metrics tags are fuzzy, human tagging was necessary to obtain a ground truth assessment of each recorded gesture. To do so, we prepared an online form with the 5-second videos presented in a randomized order. Annotators were asked to evaluate each video (without audio) using a scale for the three dimensions outlined in the previous section. In the processing of tags, the expert's opinion was given three times more weight than a non-expert scoring. To evaluate annotation consistency and reliability and ensure the validity of results derived from annotated data, we applied widely-used statistical measures for assessing inter-rater agreement or reliability among annotators: Cohen's Kappa [7] and Fleiss's Kappa [9]. Both coefficients quantify the extent of agreement between annotators, considering the possibility of agreement occurring by chance. Cohen's kappa is designed for situations involving two annotators, while Fleiss's kappa can be applied to situations with more than two annotators. This coefficient ranges from 1 (evaluators agree) to 0 (no agreement). The metrics were already used for gesture assessment (i.e., [13]).

2.3 Preprocessing and Parameterization

We normalized the recorded data to $[-1, 1]$ range. The files were then concatenated along the Z axis using the `dstack` method. To obtain the feature vectors, we utilized the Time Series Feature Extraction Library (TSFEL) [2], offering a comprehensive suite for analyzing complex time series.

2.4 Classification with Neural Networks

Support Vector Machines (SVM) have demonstrated effectiveness in discriminating between different types of gestures based on extracted features from motion data [16]. Castellano et al. [4] used SVMs for emotion recognition based on body movement and gesture dynamics, achieving an average recognition rate of 74.3%.

Long Short-Term Memory (LSTM) networks have been effective in modeling temporal dependencies, making them suitable for gesture recognition. For instance, Baccouche et al. [1] applied LSTMs to recognize human actions in videos, illustrating their ability to capture and learn relevant temporal patterns in gestures and body movements. The authors reported an accuracy of 92.9% on the KTH dataset, illustrating that LSTMs can capture and learn relevant temporal patterns in gestures and body movements.

One-dimensional convolutional neural network (1D-CNN) have demonstrated remarkable success in accurately identifying and classifying various gestures from time series data, such as an accelerometer or gyroscope signals [22]. Yang et al. [22] achieved an impressive recognition accuracy of 96.07% hand gestures classification.

Convolutional Neural Networks-Long Short-Term Memory (CNN-LSTM) have been applied to various data types, such as accelerometer, gyroscope, and depth data, demonstrating remarkable performance and robustness. For example, Nweke et al. [19] used a CNN-LSTM model to recognize hand gestures based on 3D accelerometer and gyroscope data, achieving an accuracy of 93.67%.

Bidirectional Long Short-Term Memory (BiLSTM) have demonstrated promising results in accurately identifying and classifying gestures from time series data, such as accelerometer, gyroscope, or electromyography (EMG) signals. For instance, Côté-Allard et al. [8] employed a BiLSTM model to classify hand gestures using EMG signals, reporting a recognition rate of 87.6%.

Convolutional LSTM (ConvLSTM) combines the strengths of both CNNs and LSTMs by incorporating convolutional structures within the LSTM cells [20]. This combination enables the model to learn spatial-temporal features, making it particularly useful for gesture recognition. For example, Zhu et al. [23] presented a deep learning framework called ST-ResNet for skeleton-based action

recognition that effectively captures both spatial and temporal features of human gestures. The proposed method achieved recognition accuracy of 70.3% using a cross-subject evaluation protocol and 80.0% using a cross-view evaluation protocol.

3 PILOT USER STUDY

3.1 Dataset

For the initial study, we recorded a single, 25-year-old male participant performing 162 sessions of gesticulation during public speaking training, with a script balancing different qualities combinations. From each session, a 5-seconds long window was selected for further processing. The corpus was set to provide two examples of each possible combination of the three variants of the gesture's metrics (i.e., dynamics, range, and distance between hands) intended to be performed by the user, hereafter referred to as auto-annotation.

Next, we asked 10 annotators to tag the recordings, including a public speaking trainer and a psychologist, while the others had no domain expertise. The reported task completion time was approximately 40-50 minutes. For Fleiss' Kappa, we obtained values of 0.428 for dynamics, 0.349 for range, and 0.377 for distance between hands, with a mean value of 0.390. The results for Cohen's Kappa are presented in Fig. 2. The percentage match between the intention of the actor and the assessed tag is presented for each annotator in Fig. 3. The overall match for all annotators was 63.77%.

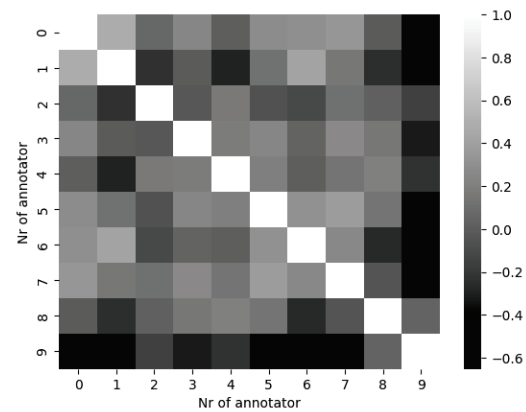


Figure 2: Cohen kappa for 10 evaluators.



Figure 3: Agreement rate between evaluators and auto-annotation.

3.2 Approach I

In the first approach, we used all recordings as training and testing data. Using the Keras library [6], we evaluated four variants of SVM (i.e., linear, polynomial, RBF, and sigmoid kernels) and all five variants of deep neural networks (DNN). The LSTM model had a hidden, single LSTM layer, a Dropout layer to reduce redundant model matching, and a dense layer combined to interpret features. The optimal number of epochs was established at 15. In the case of CNN1D, the features were reduced to a one-dimensional vector after pooling and passed through the combined layer before the output layer used for forecasting. Moreover, we used 64 parallel feature maps, and the kernel size was 3. The data set was randomly split between training and testing sets in 80:20% proportion. We also performed 10-fold cross-validation. The results are presented in Tab. 1 and Tab. 2.

Table 1: SVM for gesture qualities classification for original data.

Gesture Quality Metric	Kernel			
	Linear	Polynomial	RBF	Sigmoid
Dynamics	60.4	70.8	72.9	71.9
Range	70.8	77.1	75.0	75.0
Distance	66.7	68.7	71.9	71.9

Table 2: DNNs for gesture qualities classification for original data.

Gesture Quality Metric	Deep Neural Network (DNN)				
	LSTM	CNN-LSTM	Bi LSTM	Conv LSTM	CNN 1D
Dynamics	60.0	62.5	57.9	67.2	61.6
Range	61.6	65.3	62.2	67.5	65.6
Distance	56.6	67.8	61.3	70.9	63.8

3.3 Approach II

In another approach, we filtered the database to include only samples, where at least 7 out of 10 annotators' opinions were consistent with the actor's intention. This resulted in three subsets: 115 examples for dynamics, 103 for range, and 108 samples for the between-hand distance. For the filtered data, Fleiss Kappa values were 0.542, 0.492, and 0.419 for dynamics, range, and between-hand distance, respectively. The average consistency between annotators and the actor was 74.7% 68.7% and 67.1%. The rest of the procedure was as in the Experiment 1. The results are presented in Tab. 3 and Tab. 4.

Table 3: SVM for gesture qualities classification for filtered data.

Gesture Quality Metric	Kernel			
	Linear	Polynomial	RBF	Sigmoid
Dynamics	68.5	82.6	82.6	82.6
Range	77.0	87.1	87.1	90.5
Distance	60.6	72.8	75.8	75.8

3.4 Approach III

In the last approach, to verify whether the window size would impact the results, each video duration was set to 1 second. For the filtered data, Fleiss Kappa was 0.444, 0.359, and 0.586 (mean 0.488). The average consistency between annotators and actor was 76.3%, 74.7%, and 78.3%, respectively. The results are presented in Tab. 5 and Tab. 6.

Table 4: DNNs for gesture qualities classification for filtered data.

Gesture Quality Metric	Deep Neural Network (DNN)				
	LSTM	CNN-LSTM	Bi LSTM	Conv LSTM	CNN 1D
Dynamics	67.4	73.9	67.8	78.3	72.2
Range	77.6	84.3	80.0	85.2	83.3
HDistance	73.6	72.3	72.7	73.6	77.7

Table 5: SVM for gesture qualities classification for 1s data.

Gesture Quality Metric	Kernel			
	Linear	Polynomial	RBF	Sigmoid
Dynamics	68.42	73.68	78.95	73.68
Range	71.43	78.57	78.57	78.57
Distance	73.68	78.57	82.14	78.95

Table 6: DNNs for gesture qualities classification for 1s data.

Gesture Quality Metric	Deep Neural Network (DNN)				
	LSTM	CNN-LSTM	Bi LSTM	Conv LSTM	CNN 1D
Dynamics	66.8	67.9	65.4	73.7	71.1
Range	69.0	72.1	70.2	75.3	72.9
Distance	65.6	72.9	67.9	74.5	71.3

4 DISCUSSION AND CONCLUSION

This study presents an initial step towards evaluating selected aspects of gesture styles using classical VR setup to capture position and acceleration data. The primary outcomes include establishing baseline results, identifying the best-performing classifiers for each quality, and comparing these results to human annotations.

Regarding gesture dynamics, human annotators assessed intensity on a three-point ordinal scale, achieving a Fleiss Kappa of 0.428 for 5-second video and 0.444 for 1-second samples. This indicates a moderate level of agreement. The best result for the original dataset comprising 5-second samples was achieved using an SVM with an RBF kernel (72.9%). In the filtered subset with agreement above 0.7, an accuracy of 82.6% was obtained using an SVM with a Polynomial, RBF, or Sigmoid kernel.

For the gesture range, human annotators assessed intensity on a three-point ordinal scale, achieving a Fleiss Kappa of 0.349 for 5-second video samples and 0.359 for 1-second samples, which indicates a fair level of agreement. The best result was obtained in the original dataset containing 5-second samples using an SVM with a Polynomial kernel (77.1% accuracy). For the filtered subset with agreement above 0.7, an accuracy of 90.5% was achieved using an SVM with a Sigmoid kernel.

Regarding gesture between-hand distance, human annotators assessed intensity on a three-point ordinal scale and reached a Fleiss Kappa of 0.377 for 5-second video samples and 0.586 for 1-second samples, reflecting moderate agreement. The best result for the original dataset comprising 5-second samples was achieved using an SVM with RBF and Sigmoid kernels (71.9%). In the filtered subset (with agreement above 0.7), an accuracy of 77.7% was obtained using a 1D CNN.

In the case of quality, human assessment was more consistent for shorter samples. The most significant improvement of Fleiss Kappa was observed for between-hand distance (+0.209), slightly lower for dynamics of gestures (+0.016) and range of gestures (+0.010). Results for machine learning algorithms were comparable for 1s and 5 seconds, which can imply that the 1-second window contains

enough information for the automatic classification. In most cases, the best performance was achieved using SVM with RBF or Sigmoid kernels. Only in the 5-second-filtered case did the 1D CNN demonstrate slightly better performance.

Future research will aim to replicate the study with a larger and more diverse sample (multiple speakers of different ages and backgrounds) to enhance the external validity of the findings. Different speakers may influence the measurements based on their gender or personality; for example, a shy person may tend to keep hands close when performing various gestures [15]. Gestures style can also depend on the social context and culture [3].

Also, additional features like head position and gaze captured by VR headset might be helpful to study speaker communication with audience.

The automatic gesture detection in virtual reality (VR) has numerous applications, such as enhancing user interaction, immersive gaming, physical therapy and rehabilitation, professional training, architectural design, virtual meetings and communication, artistic expression, emotion recognition, and accessibility support for individuals with disabilities.

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